CYB102 | Intermediate Cybersecurity

Intermediate Cybersecurity Fall 2026 (@ Section 3 | Wednesdays 5PM - 7PM PT)
Personal Member ID#: 92542

Need help? Post on our class slack channel or email us at support@codepath.org

Getting Started

Learning with AI ✨

▶ Course Information

Course Progress

▼ Individual Projects

Unit 1

Unit 2

Unit 3

Unit 4

Unit 5

Unit 6

Unit 7

▶ Capstone Project

Schedule

Overview | Lab | Project | Resources | Report

📬 Submit this assignment by **Monday, October 5th at 2:59AM EDT** using the Submit

Submit button 👉

Unit 3: Project 3 - Off Limits

This project is **Part 3** of the open-source Snort NIDS Lab, built by CodePath. In Parts 0–2 you learned to run Snort and write detection rules. Now you'll run the full SOC loop on your own: exploit a real vulnerability, then detect and scope it.

No Azure Labs or VM needed — everything runs in the Docker container from your Lab.

Overview

You're a red-teamer assessing **Wishful Thinking Inc.** They run an old FTP service that hands back whatever file path a client asks for and never checks it — a **directory-traversal** vulnerability. First you'll exploit it to reach files you were never meant to see; then you'll flip to the blue team and use Snort + packet analysis to **detect the attack and prove exactly what it reached.**

👉 **View Part 3 Instructions on GitHub** 👈

🎯 Goals

By the end of this project you will be able to...

- [ ] Exploit a directory-traversal bug to read files outside a service's folder
- [ ] Write a Snort rule that detects the traversal on the wire
- [ ] Scope a breach from a packet capture — prove which files were reached
- [ ] Recommend the fix that closes the vulnerability

For this assignment, you'll be filling and submitting a copy of the 📄 **Project 3 Submission Template** (Google Doc)

- Before proceeding, we recommend you **open it up now** and read over the requirements in the document.
- It might be easier to "fill-as-you-go" than try to fill it all out after you complete the project.

▼ Required Challenges

To receive full credit, you must...

- [ ] **Recover the off-limits files via the traversal.** Include the text you pulled back for each:
  - the host's OS and version (from `/etc/os-release`)
  - the first (`root:`) line of `/etc/shadow`
  - the real earnings figure from the confidential file your account was denied
- [ ] **Write a Snort rule that detects the traversal** and add it to `local.rules`. Include the rule line and the `alert_fast` output showing it **fires four times** on `server.pcapng`.
- [ ] **Scope the breach.** From the capture, list the files the attacker actually reached and mark the ones *outside* the FTP folder. (A complete scope: 6 served requests, **4** of them `../` traversal hits.)
- [ ] **Write a remediation recommendation** (a sentence or two): the single server-side change that stops the attack at the source, and a separate defense-in-depth measure that would have limited the damage.

Close Section

▼ Stretch Challenge

To receive bonus points, you can...

- [ ] **Beat your own rule** (Part 3 stretch): send the traversal **URL-encoded** (`%2e%2e%2f`) so your literal `../` rule misses it, then write a second rule that catches the encoded form too. Include both the bypass and the new rule.

Close Section

▼ 💡 Hints

Stuck? See the **Tips for Success** at the bottom of Part 3 on GitHub — it covers finding alert lines, syntax errors that make Snort quit with 0 alerts, and pointing `-r` at the right capture.

Close Section

📬 Submitting Your Project

- 📄 **Project 3 Submission Template** (Google Doc)

▼ ✔ Am I Ready to Submit?

Check if you're **ready to submit** with the following questions:

- [ ] Did you complete all of the **Required Challenges**?
- [ ] Did you copy and fill out the [Project 3 Submission Template]?
    - It is important that you follow the same layout as the Google doc template so that we can easily access your work.
    - Be sure to check off each feature that is implemented in the **"Submission Checklist"** section
- [ ] Are any required images/GIFs/videos correctly displaying in your document?
- [ ] Did you set your document to **"*Anyone with the link can Edit*"**?

If you answered **yes** to **all** of these questions, you are ready to submit!

Look for the **"Submit"** button at the top of this page.

Close Section

Submissions Policy

- All students are allowed deadline extensions of up to 48 hours on assignments.
- Students do not need to submit a request to use these extensions; they will be automatically applied.
- Students must submit all assignments before the deadline or deadline extension in order to stay actively enrolled in the course.
- **Students are not allowed to have any missing assignments at any point during the course.**